

RAHUL JAGTAP

Data Engineer / Data Analyst | Python, PySpark & SQL | AI / GenAI & LLM Integration

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PROFESSIONAL SUMMARY

Data Engineer / Data Analyst with **4+ years of experience** designing, automating, and maintaining large-scale **ETL pipelines** and delivering analytics on multi-million record datasets. Strong hands-on **Python** and **PySpark** development, **Spark SQL** for large-scale querying, and **SQL** across **Microsoft SQL Server (T-SQL)**, PostgreSQL, and MySQL. Skilled in **data modeling**, **data profiling**, **source-to-target mapping**, and rigorous **data validation and quality control**, with hands-on **Generative AI (GenAI)** application — local LLMs (Ollama, vLLM), prompt engineering, and OCR/NER to unlock unstructured data. Comfortable in **Linux** with shell scripting and disciplined in **Git / GitLab, CI/CD, and code reviews**. Believes documentation is as important as code. Twice awarded Employee of the Month for pipeline automation and data accuracy. Open to occasional business travel.

CORE COMPETENCIES

ETL Pipeline Design • Data Engineering • Python & PySpark • Spark SQL • SQL / T-SQL • Data Modeling • Data Profiling • Source-to-Target Mapping • Data Transformation & Pipelining • Data Quality & Validation • Unstructured & Semi-Structured Data • Distributed Processing • Generative AI (GenAI) & LLM Integration • Prompt Engineering • Data Analytics & Business Reporting • Linux & Shell Scripting • Git / GitLab & CI/CD • Technical Documentation • Agile / Scrum

TECHNICAL SKILLS

Languages & Scripting: Python (Pandas, NumPy, PySpark), SQL / Standard SQL, Spark SQL, Linux / Shell Scripting

Databases & RDBMS: Microsoft SQL Server (T-SQL, Views, Query Optimization, Performance Tuning), PostgreSQL, MySQL, Database Design Principles

ETL & Data Engineering: Enterprise ETL Pipeline Design, Data Modeling, Data Profiling, Source-to-Target Mapping, Data Transformation, Incremental Processing, Job Scheduling & Orchestration, REST API Integration

Big Data & Distributed Processing: PySpark (DataFrame API, Transformations, Lazy Evaluation, Incremental Processing), Spark SQL for large-scale querying, Multi-million Record Datasets

AI / GenAI & LLM: Generative AI Applications, Local LLM Deployment (Ollama, vLLM), Prompt Engineering, LLM-based Information Extraction, LLM Inference Optimization, spaCy NER, Tesseract OCR, PaddleOCR

Cloud & Storage: AWS S3, Analytical Data Stores

Data Quality & Testing: Multi-stage Data Validation, Quality Checks, Data Testing, Root Cause Analysis, Data Accuracy & Consistency

Data Analytics: Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis, Business Reporting, Insight Generation, Stakeholder Communication

Version Control & Workflow: Git, GitHub, GitLab, CI/CD Workflows, Code Reviews, Branching & Pull Requests, Agile / Scrum, Jupyter Notebook, Excel (formulas, pivot tables)

Data Extraction: Scrapy, Selenium (structured & semi-structured data sources)

PROFESSIONAL EXPERIENCE

Data Analyst / Data Engineer — ALTR Software Services Ltd., Pune

June 2024 – Present

- Independently designed, implemented, automated, and maintained large-scale **ETL pipelines** in **Python and SQL**, processing millions of records from databases, REST APIs, and raw files (CSV, JSON, Excel) into analytical databases and **AWS S3**.
- Built incremental data processing frameworks using **PySpark** (DataFrame API, transformations, lazy evaluation) and used **Spark SQL** for large-scale querying — eliminating redundant computations on multi-million record datasets and reducing pipeline run times.
- Built and optimized queries and data models in **Microsoft SQL Server (T-SQL)** and PostgreSQL — designing target schemas, views, and transformations to support analytical and reporting layers.
- Performed **data profiling** across source systems and defined **source-to-target mappings** to drive transformation logic, assessing completeness, structure, and quality before ingestion.
- Designed a multilingual **Generative AI (GenAI) extraction pipeline** using local LLMs served via **Ollama and vLLM**, with **prompt engineering**, Tesseract OCR, and spaCy NER — parsing unstructured real estate documents and replacing manual extraction workflows entirely.
- Deployed and optimized LLM inference using **vLLM** for high-throughput batched processing of large document volumes, improving extraction speed and resource efficiency.
- Executed multi-stage **data validation, quality checks, and root cause analysis** to ensure accuracy, consistency, and reliability across all data outputs.
- Performed **exploratory data analysis (EDA)**, data cleaning, and statistical analysis, delivering business reports and analytical summaries that translated complex data into **decision-ready insights** for management and cross-functional stakeholders.
- Automated job scheduling, logging, and error handling on **Linux** using shell scripting, reducing manual intervention and improving operational consistency.

- Managed code in **Git / GitLab** with structured branching, pull requests, **code reviews**, and **CI/CD** workflows; authored **data dictionaries, pipeline runbooks, and mapping documents**.
- **Awarded Employee of the Month** for outstanding contributions to data extraction efficiency and pipeline automation.

Python Developer — Realisieren Technology Pvt. Ltd., Pune

February 2022 – June 2023

- Designed, tuned, and optimized SQL queries in **Microsoft SQL Server** and PostgreSQL for storage, retrieval, and performance tuning of multi-million-record datasets supporting downstream analytics and reporting.
- Built and maintained **ETL workflows** with job scheduling, REST API integrations, and logging on Linux, delivering clean, analysis-ready datasets to reporting teams across **E-commerce and US Oil & Gas** domains.
- Implemented **data validation and quality checks** and defined **source-to-target mappings** across multiple source systems feeding centralized reporting datasets.
- Built scalable Python-based data extraction pipelines (Scrapy, Selenium) collecting large-scale structured and semi-structured datasets feeding analytical workflows.
- Collaborated with non-technical and cross-time-zone stakeholders to scope data capabilities and communicated findings in business-friendly language.
- Managed version control in **Git** with team-level **code reviews** and collaborative workflows.
- **Awarded Employee of the Month** for excellence in process automation and data accuracy improvements.

IoT Developer Intern — Vertechxa Technology Pvt. Ltd., Pune

May 2021 – January 2022

- Developed an end-to-end IoT data pipeline using Python, Raspberry Pi, and Arduino — translating hardware sensor outputs into structured, analyzable data streams.

EDUCATION

Master of Computer Applications (MCA) — M.E. Society's Institute of Management and Career Courses, Pune Expected 2026

BBA (CA) — Bachelor of Business Administration (Computer Applications) — Maharashtra College of Science and Commerce, Pune 2015 – 2018
Completed with Higher Second Class

CERTIFICATIONS

- Google Data Analytics Professional Certificate — *In Progress* (Google / Coursera)
- Python Data Analysis — LinkedIn Learning • Python Essential Training — LinkedIn Learning • SQL for Data Analysis — LinkedIn Learning

KEY ACHIEVEMENTS

- **Employee of the Month (twice)** at two separate organizations for measurable improvements in pipeline automation, data extraction efficiency, and data accuracy.
- Built an **AI/GenAI-powered multilingual pipeline** (local LLMs via Ollama & vLLM + OCR + NER) that replaced manual document processing at scale.
- Designed incremental PySpark / Spark SQL frameworks that eliminated redundant computations on million-record datasets, improving repeatability of scheduled jobs.
- Delivered ETL, analytics, and data-validation solutions across multiple clients and domains (E-commerce, US Oil & Gas, Real Estate).